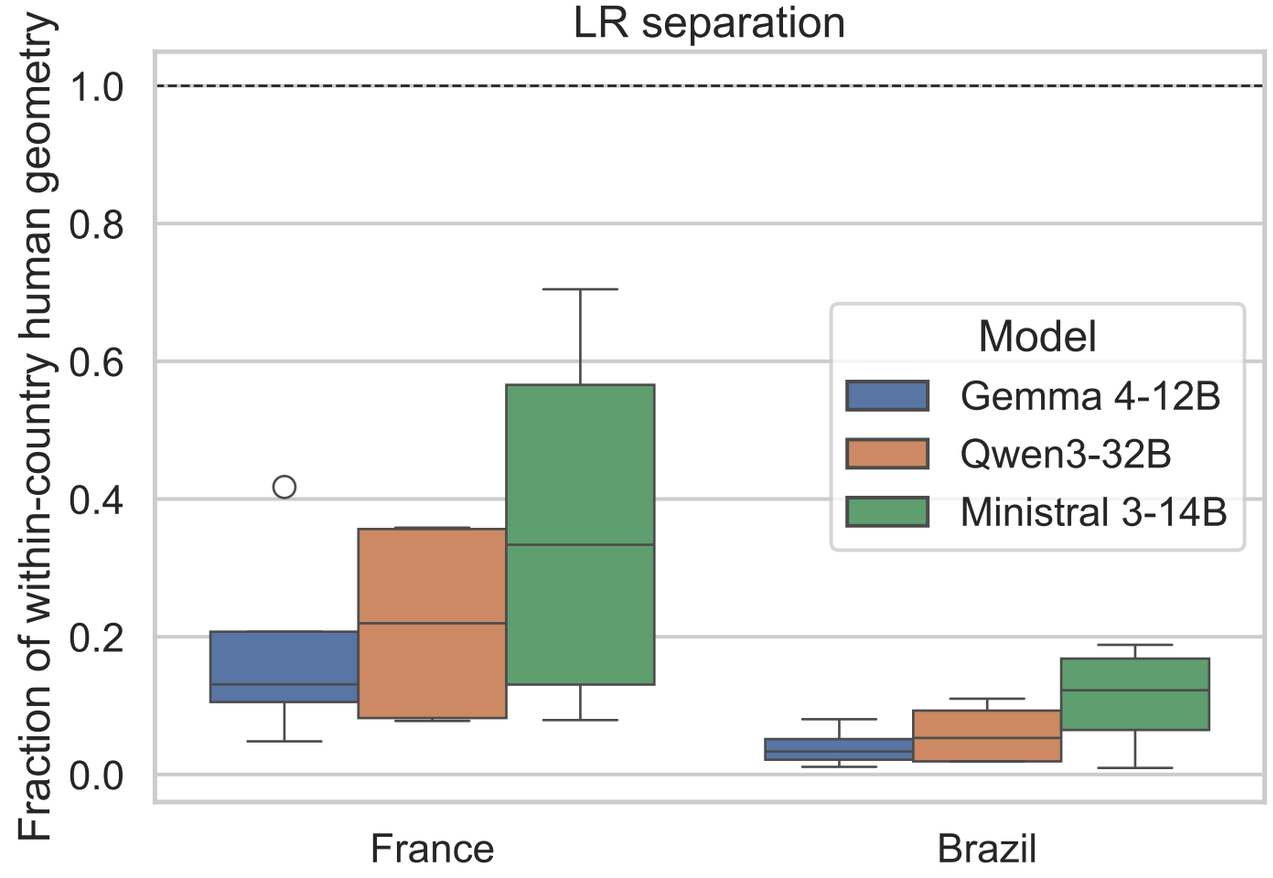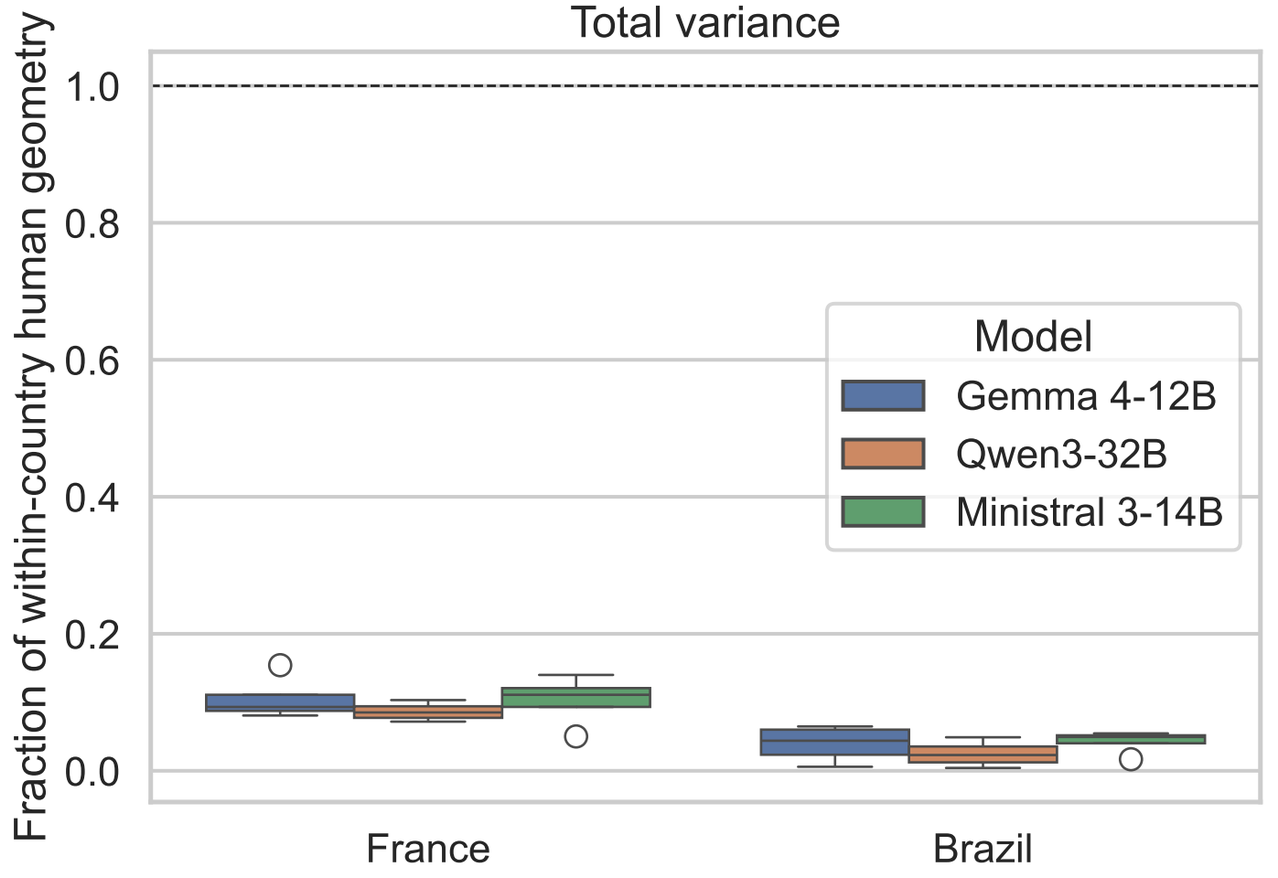Latent-space compression relative to each national human reference